ggplot2 grammar and multi-panel layouts

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Build a ggplot by naming data, aesthetics, geoms, and scales rather than by calling a plot type.
- Use faceting to display the same relationship across subgroups.
- Combine several plots into a single figure with patchwork.

Prerequisites

Labs 1.2 through 1.4.

Background

ggplot2 is built on a grammar of graphics. Rather than call a function named "scatterplot" or "boxplot", you declare what the plot contains: a dataset, mappings from variables to aesthetics (x, y, colour, size, shape), and one or more geoms that render those mappings. The result is that *any* plot you can picture can be built by the same small vocabulary. Once the vocabulary is second nature, you will spend less time searching for a plotting function and more time thinking about what the graph should show.

Multi-panel figures are the standard unit of publication. Faceting breaks one plot into small multiples — the same relationship repeated across a grouping variable — which is almost always a better answer than crowding more colours onto a single panel. The patchwork package takes independent ggplots and assembles them into a single figure using `+`, `|`, and `/` operators, so you can build a three-panel manuscript figure without leaving R.

A common beginner mistake is to treat ggplot as a collection of `geom_*` functions. It is cleaner to think of a plot as a data-plus- aesthetics object to which geoms are added. Changing the underlying data is then a substitution, not a rewrite.

Setup

```
library(tidyverse)
library(palmerpenguins)
library(patchwork)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Build a three-panel figure that describes bill morphology and body mass in the `penguins` dataset, using a scatter plot, a boxplot, and a faceted scatter plot, assembled with patchwork.

2. Approach

```
  drop_na(bill_length_mm, bill_depth_mm, body_mass_g, species, sex)
```

The plot object is built in three lines: data, aesthetics, geom.

```
p1 <- p |>
  ggplot(aes(bill_length_mm, bill_depth_mm, colour = species)) +
  geom_point(alpha = 0.7) +
  labs(x = "Bill length (mm)",
       y = "Bill depth (mm)",
       colour = NULL)
p1
```

3. Execution

```
p2 <- p |>
  ggplot(aes(species, body_mass_g, fill = species)) +
  geom_boxplot(alpha = 0.6, colour = "grey30") +
  labs(x = NULL, y = "Body mass (g)") +
  theme(legend.position = "none")
p2
```

```
p3 <- p |>
  ggplot(aes(bill_length_mm, body_mass_g, colour = sex)) +
  geom_point(alpha = 0.6) +
  facet_wrap(~ species) +
  labs(x = "Bill length (mm)",
       y = "Body mass (g)",
       colour = NULL)
p3
```

Faceting shows the same relationship in three panels, one per species. The within-species slopes are visually separable from the between-species differences.

4. Check

Assemble the three panels with patchwork.

```
(p1 | p2) / p3 +
  plot_annotation(
    title = "Palmer penguins: bill morphology and body mass",
    tag_levels = "A"
  )
```

The layout operators: `|` places side-by-side, `/` stacks, `+` groups. `plot_annotation()` adds title and the familiar A/B/C labels used in manuscripts.

5. Report

> A three-panel figure was assembled from the `palmerpenguins` data (n = `nrow(p)` complete cases) using ggplot2 and patchwork. Panel A shows a clear separation of bill morphology among the three species. Panel B shows Gentoo as markedly heavier than the other two. Panel C, faceted by species, reveals a within-species positive association between bill length and body mass, with a sex offset visible in each panel.

Within-group and between-group variation are separate features of the data. Faceting is the simplest way to let a reader see both at once.

Common pitfalls

- Using colour to encode more than one thing at a time (group *and* magnitude — pick one).
- Mapping a continuous variable to shape.
- Overlaying more than four or five categories on a single scatter plot; use faceting instead.
- Defaulting to `theme_grey()`; the in-house convention is `theme_minimal(base_size = 12)`.

Further reading

- Wickham H. *ggplot2: Elegant Graphics for Data Analysis*, chapters on the grammar and on faceting.
- Pedersen T. *patchwork* package documentation.

Session info

```
```

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)